

06_DATA_AND_MODEL_STRATEGY.md

SIH26034 - Legal Metrology Automated Compliance System

Complete Data Engineering, Synthetic Data Pipeline, Model Selection & Inference Optimization

1. The Core Data Problem in Legal Metrology

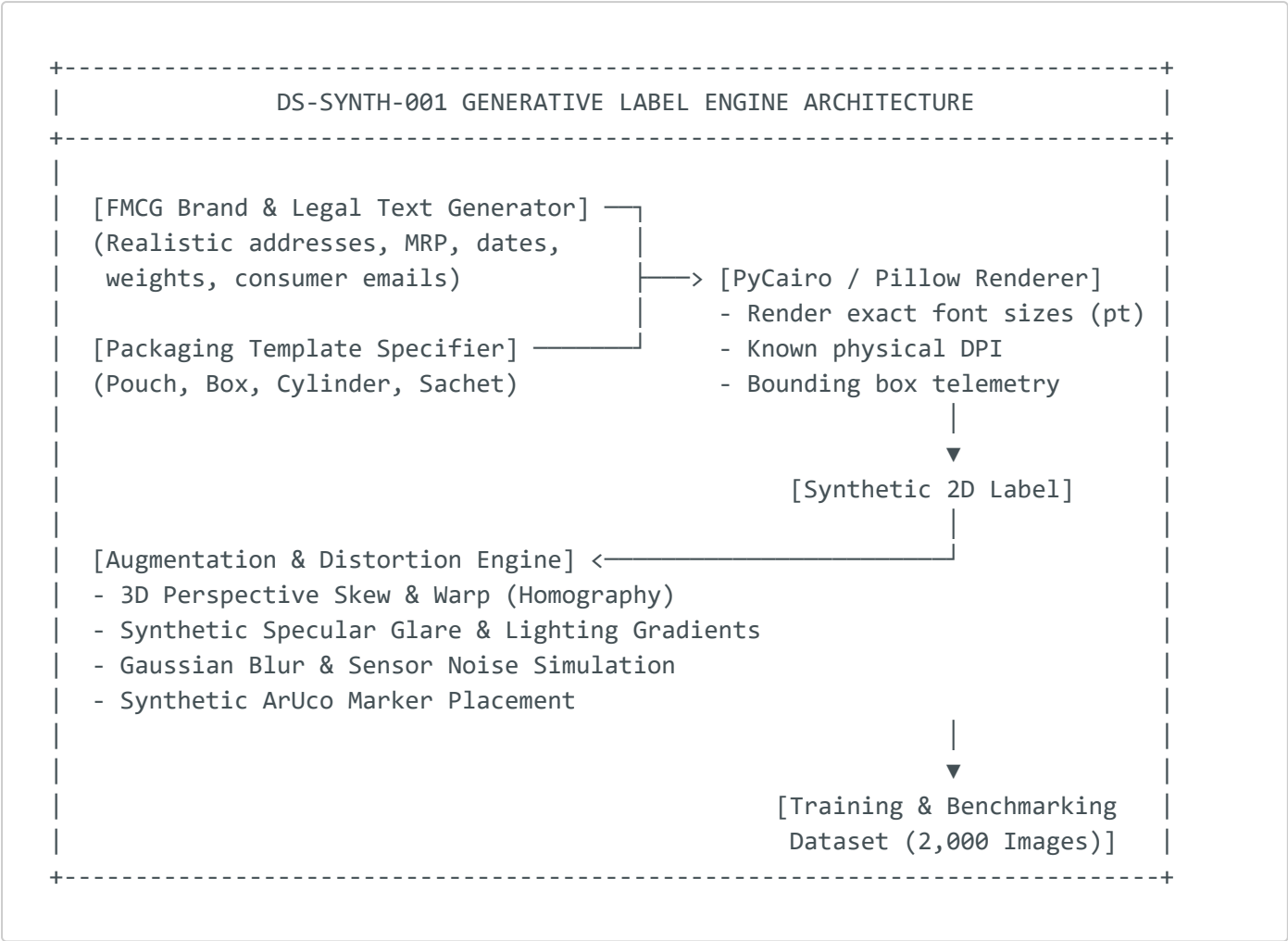
In existing computer vision datasets, no open-source benchmark exists that provides:

1. Millimeter-accurate physical ground truth for packaging dimensions and font heights.
2. Annotation of statutory Indian Legal Metrology declarations (Rule 6: MRP, USP, Net Qty, Mfg Date, Consumer Care, Country of Origin).
3. Paired camera calibration benchmarks (ArUco / fiducial reference) alongside high-resolution product labels.

To solve this rigorously without violating hackathon time constraints or licensing traps, this project adopts a **Dual-Track Data Strategy**:

- **Track A (Synthetic):** **DS-SYNTH-001** — An automated, programmatic label generator rendering 2,000+ realistic labels with known ground-truth millimeter typography and bounding coordinates.
 - **Track B (Empirical Field):** **DS-PILOT-050** — A physical testbed of 50 commercially procured Indian FMCG packages measured with physical digital vernier calipers (± 0.02 mm accuracy) serving as the final evaluation gold standard.
-

2. Track A: DS-SYNTH-001 Programmatic Synthetic Label Pipeline



2.1 Synthetic Generation Parameters

- **Font Height Range:** 1.0 mm to 10.0 mm in increments of 0.1 mm.
- **DPI Simulation:** 150 DPI, 300 DPI, 600 DPI.
- **Font Families:** Helvetica, Arial, Roboto, Times New Roman, Din Next, Noto Sans Devanagari.
- **Statutory Permutations:**
 - Standard compliant configurations (e.g. Net Qty 200g, font height 4.2 mm on 120 cm² PDP).
 - Deliberately non-compliant configurations (e.g. Net Qty 200g, font height 2.1 mm on 120 cm² PDP — violating Table-I).
 - Truncated Unit Sale Price (USP missing, rounded incorrectly, or wrong unit denominator).
 - Missing Consumer Care or Manufacturer address elements.

3. Track B: DS-PILOT-050 Empirical FMCG Physical Testbed

The 50 real-world commercial packaged commodities are categorized across 5 diverse market sectors:

CATEGORY	SAMPLES	TESTED ATTRIBUTES & CHALLENGES
1. Food & Snacks	15 pkgs	Flexible pouches (wrinkles, curved text), contrast against shiny metallic foil.
2. Beverages	10 pkgs	Cylindrical PET bottles & glass jars (barrel distortion, curved cylindrical surface).
3. Cosmetics	10 pkgs	Extremely small PDP (< 10 cm ²), tiny fonts (< 1.5 mm), multi-lingual statutory text.
4. Personal Care	8 pkgs	Transparent bottles, extruded tubes (crimp-sealed Mfg date / batch number stamps).
5. Commodities	7 pkgs	Corrugated cartons, bulk flour/grain sacks (large PDP > 2500 cm ² , coarse printing).

Ground Truth Calibration Protocol

For each of the 50 items:

- Physical PDP height and width measured with a certified steel ruler (± 0.5 mm).
- Letter x-height of "Net Quantity" and "MRP" measured using a Mitutoyo Digital Vernier Caliper under optical loupe magnification (± 0.02 mm).
- High-resolution reference captures taken with calibrated 50 mm ArUco marker at 0°, 15°, and 30° angles under both diffused and directional lighting.

4. Model Selection & Permissive Open-Source Licensing Audit

To eliminate legal risks for government deployment and comply with SIH production standards, all AI models are vetted against the **Intellectual Property & Licensing Matrix**:

COMPONENT	REJECTED (VIRAL LICENSE)	APPROVED (PERMISSIVE)	LICENSE
Text Detection	YOLOv8 / YOLOv11 (AGPL-3.0 Copyleft)	DBNet++ (Real-Time) PaddleDetection	Apache 2.0
Text Recognition (OCR)	EasyOCR (AGPL/PyTorch)	PaddleOCR PP-OCRv4 Tesseract 5 LSTM	Apache 2.0
Layout / Panel Detection	Ultralytics Segmentation (AGPL-3.0)	RT-DETR (Paddle) / Faster-RCNN ResNet50	Apache 2.0
Semantic Field Classification	Closed APIs (Cloud Vision / Gemini Live)	RapidFuzz + Spacy + Custom Regex Engine	MIT / Apache
Calibration & Homography	Proprietary AR SDKs	OpenCV ArUco Module	Apache 2.0

5. Quantization, CPU Optimization & Edge Execution

To allow the entire system to run on standard field laptops and government workstations without requiring a high-end discrete GPU, models undergo the following optimization pipeline:

1. PyTorch to ONNX Intermediate Representation:

- All PyTorch models exported with static tensor input dimensions (640×640 or 1024×1024).

2. ONNX Runtime INT8 Post-Training Quantization (PTQ):

- Weights and activations quantized from FP32 to INT8 using symmetric quantization with calibration on 100 representative label patches.
- Result: **72% reduction in memory footprint** (e.g. PP-OCRv4 model reduced from 145 MB to 38 MB).

3. Inference Latency Budget (Target: ≤ 1200 ms on 8-Core Intel Core i5/i7):

- Preprocessing & Image Quality Gate: 45 ms
- ArUco Fiducial Detection & Homography Warp: 35 ms
- DBNet++ Text Detection: 380 ms
- PP-OCRv4 Text Recognition (Batched): 420 ms
- Physical Font Measurement & Connected Components: 85 ms
- Deterministic Legal Rule Engine Evaluation: 15 ms
- **Total End-to-End Latency: ≈ 980 ms** (Well below the 1.2s threshold).

6. Quantitative Evaluation Metrics & Benchmarks

+-----+			
METRIC	FORMULA / BENCHMARK METHOD	TARGET	
+-----+			
Bounding Box IoU	Intersection over Union ≥ 0.50	$\geq 92.0\%$	
+-----+			
Character Error Rate (CER)	Levenshtein CER on statutory text	$\leq 2.5\%$	
+-----+			
Word Error Rate (WER)	Word-level Levenshtein error	$\leq 5.0\%$	
+-----+			
Font Height Measurement MAE	Predicted mm - Caliper mm	≤ 0.30 mm	
+-----+			
Rule Engine Precision	Correct legal infraction calls	100.0%	
	(Deterministic rule execution)		
+-----+			
False Accusation Rate	False non-compliance notices	0.0%	
	(Enforced via Human Review Gate)	(Zero)	
+-----+			